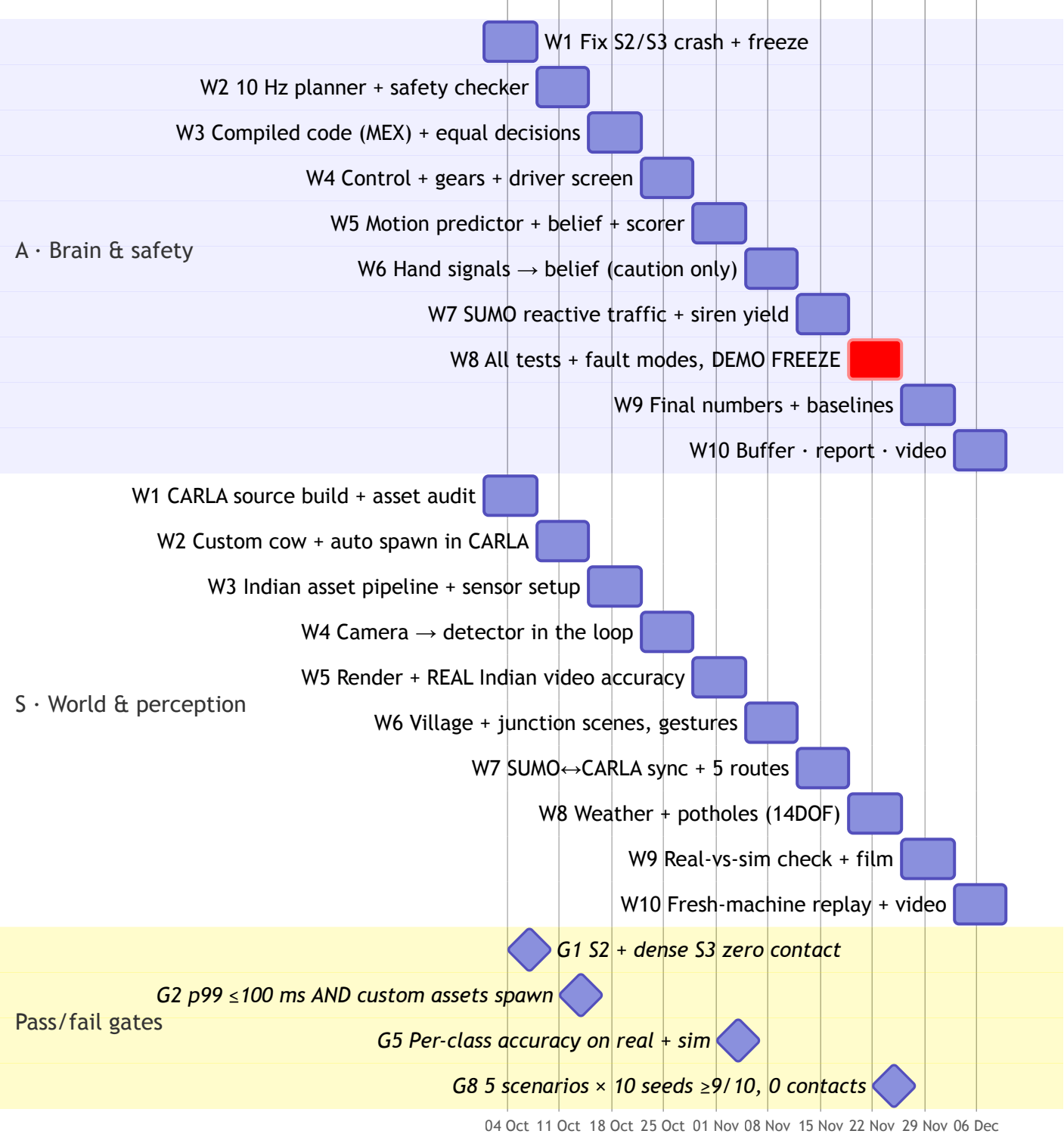


D5 · 10-week build plan with pass/fail gates

A = Aditya (brain, safety, evaluation) · S = Shourya (world, assets, perception). Both merge into ONE Simulink model every week.

10-week build plan · from 1 Oct 2026



D5 · 10-week build plan with pass/fail gates — exact spec for redrawing

Redraw rule for Gemini / Nano Banana: keep every box, group and arrow listed. Do not add, merge or drop any. Styling is free; structure is fixed.

ROWS (two parallel tracks, one exit test per week):
Week | A · Aditya: brain, safety, evaluation | S · Shourya: world, assets, perception | Exit test
W1 1–7 Oct | Fix S2 crash/freeze and dense S3 freeze | CARLA source build (165 GB, UE4.26, VS2019), asset audit | S2 + dense S3: zero contact, no false stall >10 s, 3 seeds each
W2 8–14 Oct | Remove 88% hotspot; 8–16 moves; safety checker; BLOCKED state | Spawn 1 custom prop (cow) + 1 custom vehicle (auto) | p99 ≤100 ms over 1,000 ticks; 100 unsafe moves all vetoed; assets spawn (else declared stand-ins)
W3 15–21 Oct | Versioned message contract; planner+checker compiled to MEX | Indian asset pipeline; sensor calibration; bridge smoke test | 5-min S1 run in one model; MATLAB = MEX decisions
W4 22–28 Oct | Localisation, Stanley+PI control, gears, driver screen, fault path | Fix YOLOX zero-detections; camera+LiDAR+radar → tracks | 5 min sensor→track→certificate→steering/brake; injected loss → logged stop
W5 29 Oct–4 Nov | Motion predictor, belief, move scorer (yield model stays off) | Label renders; run REAL Indian frames + own dashcam clip | Per-class accuracy on sim AND real + the gap; ML can never bypass checker
W6 5–11 Nov | Hand-signal → belief/right-of-way (caution only) | Village + junction CARLA scenes; walker raised-hand poses | All stop gestures slow/stop; zero 'go' gestures cause an uncertified move
W7 12–18 Nov | SUMO Client/Reader/Writer, reactive belief, siren yield, yield-rate sweep | Mirror SUMO actors into CARLA; 5 route files; audio events; cattle onset | 1,000 synced ticks; our move changes their move; siren-yield passes; 1 run per scenario
W8 19–25 Nov | Top-25 + 6 communication probes; fault modes; full-loop timing; DEMO FREEZE | F1–F10 mixes; rain/fog/night; 14DOF potholes; replay one mined failure | 5 scenarios × 10 seeds: ≥9/10 complete, 0 contacts; p50/p95/p99 published
W9 26 Nov–2 Dec | Frozen holdout, baselines, PS metrics, confidence intervals | Real-vs-sim recheck; film matched runs | All metrics by scenario; 299-seed bound only if all 299 clean
W10 3–9 Dec | Fix blockers; report; 8-min script | Fresh-machine replay; final video | One command reproduces every run ID and metric
CUT ORDER if late: asset variety → rear/side cameras + driver watcher → audio detail → 14DOF breadth → auto failure mining → 299-run claim.
NEVER CUT: one Simulink model · 5 scenarios · camera+LiDAR+radar loop · safety checker · safe braking · honest timing · 3 PS metrics.

Mermaid source (paste into mermaid.live to re-render)

```
gantt
title 10-week build plan · from 1 Oct 2026
dateFormat YYYY-MM-DD
todayMarker off
axisFormat %d %b
section A · Brain & safety
W1 Fix S2/S3 crash + freeze :a1, 2026-10-01, 7d
W2 10 Hz planner + safety checker :a2, after a1, 7d
W3 Compiled code (MEX) + equal decisions :a3, after a2, 7d
W4 Control + gears + driver screen :a4, after a3, 7d
W5 Motion predictor + belief + scorer :a5, after a4, 7d
W6 Hand signals → belief (caution only) :a6, after a5, 7d
W7 SUMO reactive traffic + siren yield :a7, after a6, 7d
W8 All tests + fault modes, DEMO FREEZE :crit, a8, after a7, 7d
W9 Final numbers + baselines :a9, after a8, 7d
W10 Buffer · report · video :a10, after a9, 7d
section S · World & perception
W1 CARLA source build + asset audit :b1, 2026-10-01, 7d
W2 Custom cow + auto spawn in CARLA :b2, after b1, 7d
W3 Indian asset pipeline + sensor setup :b3, after b2, 7d
W4 Camera → detector in the loop :b4, after b3, 7d
W5 Render + REAL Indian video accuracy :b5, after b4, 7d
W6 Village + junction scenes, gestures :b6, after b5, 7d
W7 SUMO→CARLA sync + 5 routes :b7, after b6, 7d
W8 Weather + potholes (14DOF) :b8, after b7, 7d
W9 Real-vs-sim check + film :b9, after b8, 7d
W10 Fresh-machine replay + video :b10, after b9, 7d
section Pass/fail gates
G1 S2 + dense S3 zero contact :milestone, g1, 2026-10-07, 0d
G2 p99 ≤100 ms AND custom assets spawn :milestone, g2, 2026-10-14, 0d
G5 Per-class accuracy on real + sim :milestone, g5, 2026-11-04, 0d
G8 5 scenarios × 10 seeds ≥9/10, 0 contacts :milestone, g8, 2026-11-25, 0d
```